

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2022
CSE-1102: Discrete Mathematics

Total Marks: 70

Time: 3 Hours

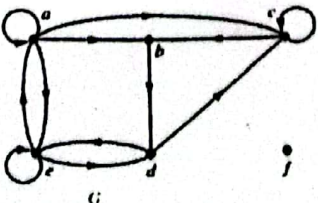
(Answer any 5 (Five) of the following Questions)

1. a) Define Tautology. Show that $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$ is a tautology by using truth table. 3
 b) Define Contradiction. Show that $(p \wedge q) \wedge \neg(p \vee q)$ is a contradiction by using truth table. 3
 c) Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences. 5
 d) Determine which rule of inference is used in the following arguments: 3
 If it rains today, then we will not have a barbecue today. If we do not have a barbecue today, then we will have a barbecue tomorrow.
 Therefore, if it rains today, then we will have a barbecue tomorrow.
2. a) What is recursion? A function f is defined recursively by $f(0)=3, f(n+1)=2f(n)+3$; find the value of $f(4)$. 4
 b) What is recurrence relation? Obtain the recurrence relation for: 5

$$G(k)=2.4^k-5.(-3)^k$$

 c) Find the solution to the recurrence relation $a_n=6a_{n-1}-11a_{n-2}+6a_{n-3}$ with the initial conditions $a_0=2, a_1=5$ and $a_2=15$ 5
3. a) Define reflexive, symmetric and transitive relation. 3
 b) Consider the following five relations on the set $A = \{1, 2, 3, 4\}$: 6
 $R = \{(1, 1), (1, 2), (2, 3), (1, 3), (4, 4)\}$
 $S = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$
 $T = \{(1, 3), (2, 1)\}$
 $U = A \times A$, the universal relation
 $V = \emptyset$, the empty relation
 Determine whether or not each of the above relations on A is
 i) Reflexive
 ii) Symmetric
 iii) Transitive
 c) Let $A = \{1, 2, 3, 4, 5, 6\}$ and let R be the relation on the set A ; 2+3
 $R = \{(x, y) \mid x \text{ divides } y\}$
 i. Write R as a set of ordered pairs.
 ii. Draw its directed graph.
4. a) Define graph. Draw the graph G , Corresponding to the adjacency matrix: 1+4

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$

 b) Define Euler circuit and Euler path in a graph. Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in figure below: 2+2

- c) Define tree and binary tree with example. Differentiate between trees and graphs. 2+3

5. a) Consider Universal set U and two sets A and B . Shade the following sets using Venn diagram: i. $A \cap B^c$ ii. $(B/A)^c$ 4
- b) f, g, h are functions on $X = \{1, 2, 3\}$ as 3
- $f = \{(1, 2), (2, 3), (3, 1)\}$
 $g = \{(1, 2), (2, 1), (3, 3)\}$
 $h = \{(1, 1), (2, 2), (3, 1)\}$
 Compute i) $f \circ g \circ h$ ii) $f \circ h \circ g$
- c) What is isomorphism? Draw two graphs that are isomorphic and describe briefly. 3
- d) In a class of 80 students, 50 students know English, 55 students know French and 46 students know German language. 37 students know English and French, 28 students know French and German, 7 students know none of the languages. How many students know all the three languages. 4
6. a) Define set with example. Write down the laws of set theory. 2+3
- b) Define one-to-one and onto function with example. Let f and g be the functions from the set of integers defined by $f(x) = 2x+3$ and $g(x) = 3x+2$. what is the composition of f and g ? what is the composition of g and f ? 2+3
- c) Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset." 4
7. a) Define Equivalence relation with example. 2
- b) Given $A = \{1, 2, 3, 4\}$ and $B = \{x, y, z\}$; let R be the following relation from A to B : 2
- $R = \{(1, y), (1, z), (3, y), (4, x), (4, z)\}$
 Determine the matrix of the relation.
- c) Let $A = \{1, 2, 3, 4\}$, $B = \{4, 5, 6\}$ and $C = \{6, 7, 8\}$; R is a relation from A to B and S is a relation from B to C , which is given by 3
- $R = \{(x, y): x + y = 7\}$
 $S = \{(x, y): y - x = 1\}$
 Determine R and S .
- d) Let $A = \{1, 2, 3\}$, $B = \{a, b, c\}$ and $C = \{x, y, z\}$; Consider the following relations R and S from A to B and from B to C , respectively. 3
- $R = \{(1, b), (2, a), (2, c)\}$
 $S = \{(a, y), (b, x), (c, y), (c, z)\}$
 Find the composition relation $R \circ S$ with arrow diagram.
- e) Find all spanning trees of the following graph: 4

